

CHARA, Russian Federation

WMO: 303720

Lat: 56.900N

Lon: 118.267E

Elev: 2331

StdP: 13.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.2	-44.9	-52.3	0.3	-47.7	-49.0	0.3	-44.8	14.5	0.4	12.1	-4.4	0.3	270	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	82.2	62.3	78.8	61.1	75.2	60.0	65.2	76.0	63.4	74.0	61.7	71.9	5.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	89.7	68.6	59.8	83.5	66.7	57.8	77.9	65.5	31.5	75.9	30.1	73.9	28.8	71.9	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.6	11.4	DB	-52.1	87.6	4.4	2.5	-55.3	89.4	-57.9	90.8	-60.4	92.2	-63.6	94.0	(4)
(5)			WB	-52.2	68.6	4.4	2.5	-55.3	70.4	-57.9	71.8	-60.4	73.2	-63.6	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.4	-24.1	-14.8	2.2	25.0	41.9	57.3	62.0	56.5	41.9	22.2	-4.7	-22.5
	DBStd	32.35	10.96	11.69	13.11	9.39	6.98	7.05	5.53	6.20	7.23	10.60	14.27	11.31
	HDD50	11653	2299	1813	1481	749	263	20	2	18	258	861	1641	2248
	HDD65	16312	2764	2233	1946	1199	715	242	123	267	693	1326	2091	2713
	CDD50	861	0	0	0	0	13	238	374	221	15	0	0	0
	CDD65	44	0	0	0	0	0	10	29	5	0	0	0	0
	CDH74	969	0	0	0	0	5	303	491	167	3	0	0	0
	CDH80	188	0	0	0	0	0	60	106	22	0	0	0	0
Wind	WSAvg	3.4	1.3	2.1	3.6	5.5	5.9	4.5	3.7	3.5	4.0	3.6	2.3	1.3
Precipitation	PrecAvg	13.6	0.2	0.1	0.2	0.6	1.5	2.9	2.6	2.8	1.9	0.6	0.2	0.2
	PrecMax	18.9	0.6	0.4	0.9	2.0	3.5	5.4	5.4	6.7	4.0	3.2	0.9	0.5
	PrecMin	7.0	0.0	0.0	0.0	0.1	0.4	0.2	0.3	0.4	0.7	0.0	0.0	0.0
	PrecStd	2.9	0.1	0.1	0.2	0.4	0.7	1.5	1.3	1.5	0.8	0.6	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	24.1	41.4	56.3	73.5	85.8	86.9	83.0	71.5	51.8	38.1	16.8
		MCWB	9.5	19.4	33.2	41.6	53.1	61.9	64.2	62.7	54.6	41.0	32.3	14.2
	2%	DB	4.7	17.3	34.1	49.6	66.7	81.4	83.0	78.7	65.5	45.4	29.8	7.9
		MCWB	2.9	14.0	27.6	37.9	49.5	60.5	63.4	61.7	52.1	37.2	25.7	6.1
	5%	DB	-1.9	10.9	29.5	44.9	61.7	77.6	79.6	74.7	60.3	40.9	23.2	0.5
		MCWB	-3.0	8.6	24.0	35.2	46.7	59.0	62.5	60.3	49.3	33.9	20.4	-0.3
	10%	DB	-7.6	5.6	25.3	40.8	57.1	73.5	76.0	70.6	55.6	37.1	16.8	-5.8
		MCWB	-8.3	3.8	20.8	32.4	44.3	57.4	61.6	58.4	46.9	31.6	14.9	-6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	19.8	32.7	42.1	53.8	65.6	67.9	66.6	57.2	41.4	32.4	14.1
		MCDB	11.7	23.9	40.7	53.3	70.1	80.4	78.8	76.1	68.6	50.3	38.1	16.4
	2%	WB	3.0	14.0	27.6	38.6	50.7	62.4	66.0	63.7	53.6	37.7	26.0	6.0
		MCDB	4.7	17.1	33.7	48.1	64.3	76.0	76.7	73.7	61.3	44.4	29.3	7.7
	5%	WB	-3.0	8.8	24.1	35.7	47.7	60.6	64.5	61.7	50.7	34.6	20.3	-0.3
		MCDB	-2.0	11.0	29.3	44.3	59.2	73.5	74.9	71.1	58.7	39.9	23.1	0.6
	10%	WB	-8.3	3.8	20.7	32.8	45.1	58.8	63.0	59.8	47.9	31.9	14.8	-6.3
		MCDB	-7.7	5.5	25.1	39.8	55.6	70.5	73.2	68.5	54.8	36.6	16.9	-5.8
Mean Daily Temperature Range	5% DB	MDBR	20.7	29.0	31.9	23.0	20.3	23.2	22.1	22.5	19.8	19.4	20.5	17.6
		MCDBR	24.3	29.9	32.0	27.0	31.3	32.8	31.4	32.6	30.2	22.2	22.1	22.3
		MCWBR	22.9	27.0	25.7	16.4	15.9	14.1	13.8	16.5	17.0	14.3	18.1	21.0
	5% WB	MCDBR	24.3	29.7	31.3	25.2	28.0	27.6	24.7	26.0	26.8	20.7	21.9	22.1
		MCWBR	23.0	26.7	25.2	15.8	14.9	12.8	11.8	13.8	15.6	13.8	18.0	20.8
Clear-Sky Solar Irradiance	taub		0.237	0.270	0.297	0.363	0.440	0.439	0.439	0.396	0.329	0.285	0.247	0.225
	taud		2.067	2.074	2.067	1.983	1.963	2.167	2.206	2.316	2.429	2.312	2.097	2.006
	Ebn at Noon		203	245	270	266	252	255	252	256	261	242	195	169
	Edh at Noon		18	28	37	48	53	44	41	35	26	22	17	14
All-Sky Solar Radiation	RadAvg		245	548	1052	1591	1710	1663	1545	1278	857	533	298	167
	RadStd		13	27	62	108	164	123	139	100	80	36	24	11

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page